

# Amir Shirian

## Senior Machine Learning Scientist

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## PROFILE

Machine learning scientist with **6+ years** of experience solving real-world problems with machine learning approaches. I also have **7+ years** of researching and proposing new methods to achieve better performance in all my projects. As my personal advancement, I stay on top of all recent developments specially **LLM** and **generative AI** techniques/research and actively incorporate them in my projects.

## WORK EXPERIENCE

### Senior Applied AI/ML Scientist

2024 – Present

*JPMorgan Chase*

- Delivered natural language products. Designed and deployed a **multi-agent system** capable of processing documents and extracting key information aligned with stakeholder requirements.
- Led a team to develop an **AI-based handwritten signature authentication** product using a **visual-lingual hybrid approach**, extracting mandate restrictions from text, identifying relevant signatures in incoming documents, and verifying them against the correct signature list. The visual comparison model was designed from scratch using **metric learning** with **self-supervised learning**, achieving **95% accuracy** and high confidence based on stakeholder defined metrics.
- Co-developed an **agentic query management** solution based on **Langgraph** for the Custody team, enabling **intent understanding** and **tool calling** across 6+ systems to retrieve trade details, with support for both short and long-term memory to maintain context.
- Fine-tuned **Small Language Models (SLMs)** for sanction identification tasks using **parameter-efficient** methods such as **LoRA** and **QLoRA**.
- **Tools:** Langchain, LangGraph, PyTorch, CLIP, SimCLR, FastAPI, OpenAI API, Docker, OpenAI Function Calling, Azure OpenAI, LoRA/QLoRA (PEFT), Comet, AWS, ONNX.
- Mentoring intern PhD students and junior members of team.

### Senior Machine Learning Engineer, Co-Founder

2024 – Present

*Emergesound.ai*

- **Training one of the very first music foundation models:** Built one of the first music foundation models using a curated dataset of **130M songs** and **4.7B tokens**, ensuring balanced genre and style representation.
- Designed a scalable, **cloud-optimized data pipeline** and **streaming architecture** that eliminated training bottlenecks through efficient data loading and integrated benchmark datasets to ensure robustness and generalization.
- Applied **advanced LLM techniques** (**ModernBERT**, **DeepSeekv3**) across software and hardware layers to enhance training performance.
- Extended model context length using **RoPE positional encoding**, increasing it from **9k to 36k tokens** in line with modern LLM capabilities.
- Delivered cloud-optimized model architectures ready for deployment on **NVIDIA Triton**, **AWS SageMaker**, and **Bedrock**, supporting **inference-ready frameworks**.

### Senior Data Scientist

2022 – 2024

*Nokia*

- Delivered **computer vision** products.
- Antenna counting on Telecom. towers using drones. Delivered **hazard detection** for workers on the site. (Recognized as the **best solution** in the year). All the components are aligned with **MLOps** and distributed best practices like **MLflow**, **PVC**, **Horovod**, and **DeepSpeed**.
- **Face verification** app as check-in process for workers.
- Hands-on developing an **edge-device model conversion pipeline** with **quantization methods**.
- A technical member of **GenAI taskforce** at Nokia to develop **GenAI strategies** aligned with Nokia's business. Selected as nominee for the best member of MN business.
- **Side projects with Bell Labs:** Video retrieval using recent models like **VideoMAE+CLIP**. Video action recognition with **language supervision**. Optimized training by **10x** using **gradient checkpointing** and **flash attention**. Video captioner to generate labels for language supervision.

## Machine Learning Engineer

2021 – 2022

*DeepMirror*

- Providing **graph solutions** for data scarcity cases in the biomedical data (images, molecules, RNA/DNA, and antibody).
- Generate potential drugs and proteins. Achieve much **higher segmentation results** with a very low number of labelled images.
- Represent **RNA data as a graph structure** for the first time and boosted the efficacy prediction performance with a huge margin. We got a **UKRI Smart grant** for this research.

## Research Collaborator

2021

*Google AI*

- Designed **novel self-supervised tasks** and extended the idea on **heterogeneous data**.

## Machine Learning Intern

2020

*Intel AI lab*

- Designing **Learnable Graph Inception Network** for **Emotion Recognition** on different modalities.

# EDUCATION

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## PhD in Computer Science

2019 – 2022

*University of Warwick*

- Warwick Computer Science PhD students scholarship.
- Hands-On Graph Neural Networks Using Python book.
- Interspeech, WACV, ICASSP, ICME, and couple of more journal reviewer.

# SELECTED PUBLICATIONS

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1. Ahmadian, M., Shirian, Amir, "DEL: Dense Event Localization for Multi-modal Audio-Visual Understanding", *ICCV* 2025.
2. Shirian, Amir, Ahmadian, M., Somandepalli, K., and Guha, T. "Heterogeneous Graph Learning for Acoustic Event Classification." *ICASSP* (2023).
3. Shirian, Amir, Somandepalli, K., Sanchez, V., Guha, T. "Visually-aware Acoustic Event Detection using Heterogeneous Graphs." *Interspeech* (2022).
4. Shirian, Amir, Somandepalli, K., and Guha, T. "Self-Supervised Graphs for Audio Representation Learning with Limited Labeled Data." *IEEE Journal of Selected Topics in Signal Processing* (2022).
5. Shirian, Amir, Subarna T., and Guha, T. "Dynamic Emotion Modeling with Learnable Graphs and Graph Inception Network." *IEEE Transaction on Multimedia*.
6. Shirian, Amir, and Guha, T. "Compact Graph Architecture for Speech Emotion Recognition." *ICASSP* 2021.